```
!pip install boto3 awscli
```

```
Requirement already satisfied: boto3 in /usr/local/lib/python3.12/dist-packages (1.40.34)
Requirement already satisfied: awscli in /usr/local/lib/python3.12/dist-packages (1.42.34)
Requirement already satisfied: botocore<1.41.0,>=1.40.34 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.40.34)
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.12/dist-packages (from boto3) (1.0.1)
Requirement already satisfied: s3transfer<0.15.0,>=0.14.0 in /usr/local/lib/python3.12/dist-packages (from boto3) (0.14.0)
Requirement already satisfied: docutils<=0.19,>=0.18.1 in /usr/local/lib/python3.12/dist-packages (from awscli) (0.19)
Requirement already satisfied: PyYAML<6.1,>=3.10 in /usr/local/lib/python3.12/dist-packages (from awscli) (6.0.2)
Requirement already satisfied: colorama<0.4.7,>=0.2.5 in /usr/local/lib/python3.12/dist-packages (from awscli) (0.4.6)
Requirement already satisfied: rsa<4.8,>=3.1.2 in /usr/local/lib/python3.12/dist-packages (from awscli) (4.7.2)
Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /usr/local/lib/python3.12/dist-packages (from botocore<1.41.0,>=1
Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /usr/local/lib/python3.12/dist-packages (from botocore<1.41.0,>=1
Requirement already satisfied: pyasn1>=0.1.3 in /usr/local/lib/python3.12/dist-packages (from rsa<4.8,>=3.1.2->awscli) (0.6.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil<3.0.0,>=2.1->botocore<
```

```
! aws configure
```

```
AWS Access Key ID [****************CRWQ]:
AWS Secret Access Key [****************Xtmx]:
Default region name [Global]:
Default output format [None]:
```

Claude 3 Sonnet via Amazon Bedrock

```
import boto3
import json
from google.colab import userdata

# Create Bedrock runtime client
# Use credentials from Colab secrets
bedrock = boto3.client(
    "bedrock-runtime",
    region_name="us-east-1",
    aws_access_key_id=userdata.get('AWS_ACCESS_KEY_ID'),
    aws_secret_access_key=userdata.get('AWS_SECRET_ACCESS_KEY')
)

# Claude prompt format: use Human/Assistant roles
prompt = "\n\nHuman: Explain machine learning to a high school student.\n\nAssistant:"

# Call Claude Instant
response = bedrock.invoke_model(
    modelId="amazon.titan-text-lite-v1",
    body=json.dumps({
        "inputText": prompt,
        "textGenerationConfig": {
            "maxTokenCount": 300,
            "temperature": 0.7,
            "topP": 0.9,
            "stopSequences": []
        }
    }),
    contentType="application/json",
)

# Parse and print output
output = response['body'].read().decode("utf-8")
print("Titan Response:\n", json.loads(output)['results'][0]['outputText'])
```

```
Titan Response:

Sure, machine learning is a branch of artificial intelligence (AI) that enables systems to learn and improve from experience wi

One of the key challenges of machine learning is the need for large amounts of data. The algorithms require a lot of training d

Another challenge is the complexity of the algorithms. Machine learning algorithms can be extremely complex, and it can be diff

Despite these challenges, machine learning has had a significant impact on many industries. It has been used to improve health

In conclusion, machine learning is a branch of artificial intelligence that enables systems to learn and improve from experienc
```

Amazon Titan Text Model

```
import boto3
```

```
import json

# Create Bedrock runtime client
bedrock = boto3.client("bedrock-runtime", region_name="us-east-1")

input_text = "Write a short story about a robot who learns to love."

response = bedrock.invoke_model(
    modelId="amazon.titan-text-lite-v1",  # You can also use titan-text-express-v1
    body=json.dumps({
        "inputText": input_text,
        "textGenerationConfig": {
            "maxTokenCount": 200,
            "temperature": 0.7,
            "topP": 0.9,
            "stopSequences": []
        }
    }),
    contentType="application/json",
)

# Parse and print the output
output = response['body'].read().decode("utf-8")
print("Titan Response:\n", json.loads(output)['results'][0]['outputText'])
```

```
Titan Response:

Once upon a time, there was a robot named Robby. Robby was created with a mission to serve and help humans. However, he had a p

One day, Robby was assigned to a hospital to help take care of patients. At first, he was hesitant to do his job, but he soon r

At first, Robby was hesitant to express his feelings to Emily. He was afraid that she would reject him or that he would be unab

As Rob
```

Start coding or generate with AI.